Course Name: Discrete Mathematics

Course Outcome

CO1: Understand the notion of mathematical thinking and proofs to solve the problem.
CO2: Apply the basics of discrete probability and number theory to solve the real world problem.
CO3: Analyze basic discrete structures and algorithms using effectively algebraic techniques.
CO4: Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.
CO5: Analyze the validity of an argument using logical notation.
CO6: Demonstrate the basic structures of proof techniques to write and evaluate the validity of arguments.
CO7: Understand the basic principles of sets, set equalities and operations in sets.
CO8: Apply counting principles to determine probabilities.

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Mid Term Examination, Even Semester 2023-24
B.TECH.(CS/AIML/CCV/CSF/DA/IIoT), IV Sem
BCSC 1010 DISCRETE MATHEMATICS

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider the multiplication of bacteria in a controlled environment. Let a_r denote the number of bacteria there are on the r-th day. We denote the rate of growth on the r-th day to be $a_r - 2a_{r-1}$. If it is known that the rate of growth doubles every day, determine a_r, given that $a_0 = 1$?	3	CO1	E	P
2	A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them. a) How many balls must she select to be sure of having at least three balls of the same color? b) How many balls must she select to be sure of having at least three blue balls?	3	CO8	An	C
3	Let R be the relation on the set of ordered pairs of positive integers such that ((a, b), (c, d)) ∈ R if and only if ad = bc. Show that R is an equivalence relation.	3	CO7	E	P
4	How many different 11 letter words are possible using the letters of the word COMBINATION such that 2 O's come together?	3	CO8	An	C
5	Prove that $n^3 - 4n + 6$ is divisible by 3 for all $n \in N$ using Mathematical Induction.	3	CO1	C	C

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Solve the given equation using generating function $a_{n+2} - 2a_{n+1} + a_n = 2^n$, $a_0 = 2, a_1 = 1$	5	CO8	E	P
7	Which elements of the poset ({2, 4, 5, 10, 12, 20, 25}, |) are maximal, minimal, greatest element and least element? Support your answer by drawing the Hasse Diagram. Whether this poset is a lattice or not? Justify your answer.	5	CO1	E	P
8	Each student in Liberal Arts at some college has a mathematics requirement A and a science requirement B. A poll of 140 sophomore students shows that 60 completed A, 45 completed B, and 20 completed both A and B. Find the number of students who have completed: (a) At least one of A and B; (b) exactly one of A or B; (c) neither A nor B.	5	CO4	E	P